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A Dual-Channel convolution neural network for image smoke detection

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Abstract
Image smoke detection is a challenging task due to the difference of color, texture, and shape of smoke. In recent years, deep learning has greatly improved the performance of image classification and detection. In this paper, we propose a Dual-Channel Convolutional Neural Network (DC-CNN) using transfer learning for detecting smoke images. Specifically, an AlexNet network with transfer learning, used to extract generalized features, is designed on the first channel as the main framework of entire network. The second channel is a tidy convolution neural network for extracting specific and detailed features. To guarantee the robustness of the network, two channels of the network are trained separately and their features are fused in the concat layer. The experimental data sets consist of smoke images and non-smoke images, and some challenging non-smoke images are added into the data sets as a supplement. Experimental results show that the proposed method can work effectively and achieve detection rate above 99.33%.

Keywords Dual-channel convolutional neural network · Transfer learning · Image smoke detection · AlexNet

1 Introduction

Fire detection is vital for people safety and property. Traditional fire detectors are widely used because of their low cost and simple principle. Most detectors use very accurate sensors to

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detect the presence of fire by detecting particles in the air, temperature, smoke and relative humidity [1]. These sensors need to physically contact with ambient gases of combustion. If sensors are far from the source of the fire, the fire cannot be detected quickly. People discover that smoke often occurs before frame in many fires, thus researchers have diverted attention to the detection of smoke. To eliminate the potential threat of fire as early as possible, researchers begin to study the image-based smoke detection technology, which can immediately identify smoke when smoke appears in the image. Obviously, the detection of smoke images is more suitable than conventional detectors in most cases, such as buildings and forest fires. With the rapid development of digital cameras and machine vision systems, image smoke detection will be widely used.

According to the spatial and temporal characteristics of smoke video images, many researches have been presented on smoke detection: Wang et al. [27] proposed an automatic forest fire detection method based on temporal contrast differences by using the spatial characteristics of smoke. The movement characteristics of smoke can be extracted to judge whether there is smoke. Yuan et al. [30, 31] produced a fire and smoke detection method based on motion accumulation and showed the law of early fires. However, these methods involve high false alarm rate and fussy processing data.

Recently, deep learning has achieved a great success in image classification and recognition territory because it has an excellent ability of automatic learning features. Some researchers have applied deep learning to smoke detection and get satisfying results: N. James et al. [12] trained a C3D convolutional neural network for recognizing wildfire smoke. But the limitations of deep learning are that lots of manually labeled samples are required and the collection of data is very difficult and expensive. Transfer learning can effectively solve this issue by transferring the pre-training model parameter to initialize the new model. In this paper, we propose a Dual-Channel Convolution Neural Network with transfer learning (DC-CNN) for smoke detection. Specifically, DC-CNN has two channels to extract features from raw pixels of image samples. For the first channel, an AlexNet network is constructed with transfer learning, which can extract more comprehensive and generalized features without manual extraction. For the second channel, a tidy CNN is built to obtain more precise and purposeful features. The paper is distinguished by the following main contributions:

1. We propose a dual-channel network for detecting smoke. The first channel uses transfer learning on the AlexNet network as the main network structure, which can make the model more generalized and rapidly converge to a better accuracy. The second channel is presented by a tidy network for getting more targeted and comprehensive features to promote the accuracy of smoke detection.
2. We construct some challenging images data sets to evaluate the image smoke detection algorithm. Experimental results demonstrate the superiority of our algorithm compared to state-of-the-art algorithms.

The rest of this paper is organized below. Section II reviews characteristics of smoke and the transfer learning development. In Section III, the DC-CNN is presented in detail. The experimental results are described in Section IV. Finally, the section V summarizes this paper.

2 Related work

Many researchers have attempted to deal with smoke detection by focusing on characteristics of smoke, such as color, shape, and edge irregularity. Three types of characteristics of smoke have been studied. First, when something is burning, the combustion temperature, the chemical composition of the fuel and the supply of oxygen will determine the color characteristics of the smoke. In fact, the color of smoke is grayish in most instances [2, 7, 9, 21]. In other words, the three intensity values in the RGB channel of smoke are equal or similar, so the essential feature can be used to distinguish smoke [9]. Second, the complex and irregular edge feature is the substantive characteristic of smoke, which is easy to distinguish from other objects. The feature can be divided into two groups. The first group is extracted to characterize the boundary of smoke. The second group describes the edges of objects covered by smoke. In the first group, histogram of oriented gradient (HOG) [14] descriptors are extracted for smoke detection [20, 25]. The second group is the blurred edges could be observed in smoke. For example, B. Töreyin, et al. [11] used the consequent decrease in high frequency as a cue to perform smoke detection. However, this decrease in high frequency is not unique to smoke coverage and is hard to measure its extent from a single image. Third, due to the existence of thermal convection, the diffusion direction of the smoke is upward. This phenomenon can be used to extract the smoke texture features and contribute to detect smoke [9, 17, 19, 23, 28]. The texture feature of smoke can be regarded as the third distinguishable characteristic.

Recently, deep learning has shown a significant potential in computer vision area [18, 32]. CNN is a multilevel self-monitoring learning perceptron which is introduced by Fukshima [15]. Lecun developed CNN to classify numbers and identify the handwritten check numbers of LeNet-5 successfully. Ciresan [3] used CNN to implement the best performance of multi-objective recognition in the multiple image databases. In 2012, AlexNet won the ImageNet contest with its greatly improved accuracy. The success of AlexNet opens the era of deep learning and facilitates the development of other network structures. Recently, the convolution neural network is widely used in image recognition field because of its excellent effect.

Labeled data is very important in deep learning. But data collection is very difficult and expensive. Meanwhile, researchers have found that transfer learning can perform well without large data samples. Transfer learning is transferring the feature parameters of the pre-training model to the current task without large labeled data [8, 33]. Parameters of the pre-training model are obtained from the ImageNet data sets and directly transferred to a new model. Transfer learning provides a better method for cross-task and cross-domain research. For example, Wan et al. [26, 34] introduced transfer learning into cross-language opinion analysis, transferring knowledge information from English product reviews to Chinese product reviews. In sentiment analysis, Aue et al. [6] introduced transfer learning successfully into cross-domain sentiment analysis between DVD, books, and music in three different areas. Transfer learning also shows good performance in image processing. Du, et al. [5] employed the idea of transfer learning from interactive annotation technique in video. Karri, S. P. K., et al. [13] presented an algorithm for identifying retinal pathologies given retinal optical coherence tomography (OCT) images. Their approach fine-tuned a pre-trained CNN to improve its prediction capability (compared to random initialization training) and identified salient responses during prediction to understand learned filter characteristics. Tadanobu, et al. [10] proposed the method transfer learning from synthetic to real images using variational autoencoders for

robotic applications. They use limited dataset of real images while leveraging a large dataset of synthetic images using multiple variational autoencoders. It detects object positions 6 to 7 times more precisely than the baseline of directly learning from the dataset of the real images. In this paper, we utilize transfer learning to make the model converge and optimize efficiently.

3 Approach

A dual-channel convolutional neural network with transfer learning is proposed in this section. The first channel is the AlexNet network with transfer learning, as the basic structure of the network to get more low-level generalized features. Transfer learning can make feature collection more rapid and accurate. The second channel is a detail-extracted CNN to extract more targeted advanced features. Two channels are trained at the same time and the acquired features are fused to detect smoke.

3.1 AlexNet with transfer learning

As the design of CNN networks becomes deeper, the cost of training networks is increased and lots of training data are required. Transfer learning shares the feature parameters of pre-training model into the new model, so the new model dispenses with learning from scratch. Transfer learning not only solves the problem of insufficient data sets, but also accelerates the learning efficiency of the new model and improves the accuracy of smoke classification.

To reduce the difference between the two domains and improve classification performance of target domain, transfer learning is used to find common feature representation in the feature space of source and target domain. The process of transfer learning is transferring their knowledge from source tasks to target tasks. In this paper, the first step is to train a pre-training model on the ImageNet basic data sets. Second, we use transfer learning to transfer the learned feature parameters to the first channel of smoke classification network. Finally, the first channel network is trained on the smoke data set to build a new model. The block diagram is shown in Fig. 1.

AlexNet mainly composes of five convolutional layers and three fully-connected layers. The structure of AlexNet makes it perform well on detecting images, so we choose AlexNet as the basic framework of our network structure. The architecture is shown in Fig. 2. In the first

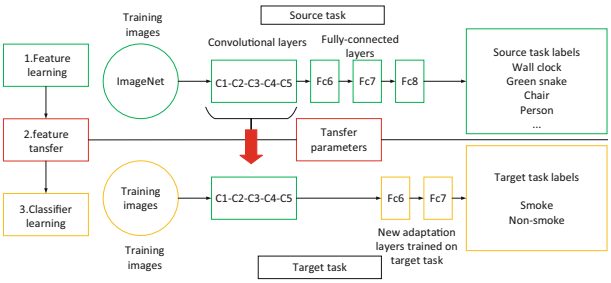

Fig. 1 Transfer learning flow chart. First, there are a large number of available labeled images (ImageNet) as the data sets to train the network of source tasks. Second, the pre-training parameters of the five-layers convolutional layer (C1-C5) are transferred to the smoke target task. Finally, to compensate for the difference between the source data sets and the smoke target data sets, data adaptation layers (fully-connected layers Fca and Fcb) are introduced

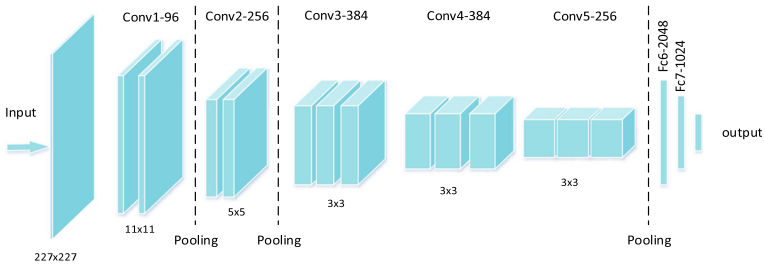

Fig. 2 The first channel: AlexNet with transfer learning

channel, there is a classic AlexNet structure for feature extraction. The weights of the first convolutional layer (Conv1) to the last pool layer (Pool5) are initialized by the pre-training model feature parameters.

The first channel network is initialized with feature parameters obtained from natural image, and the smoke features are extracted as the target image features. By using transfer learning, the characteristics of the target task can be extracted quickly and the classification accuracy can be improved.

3.2 Detail-extracted CNN

Transfer learning solves the problem of insufficient data sets. The new model obtains the parameters from the pre-training model, so the new model can quickly obtain convergence and learn excellent generalization features. Meanwhile, we need more detailed features of the target images, as the more comprehensive features will make the final classification more accurate. Therefore, we construct a tidy network based on AlexNet as the second channel. This channel is trained independently to obtain the detailed features of smoke images. The network structure is shown in Fig. 3.

In the second channel, a detail-extracted convolution neural network is used to train the object data sets directly. It can extract more specific features of smoke. The detail-extracted CNN contains four convolutional layers (Conv1_1 - Conv4_1) and one pooling layer (Pool2_1).

3.3 DC-CNN

To increase the robustness of the whole network, two channels of the network extract features independently. In the first channel, the method of transfer learning not only solves the problem

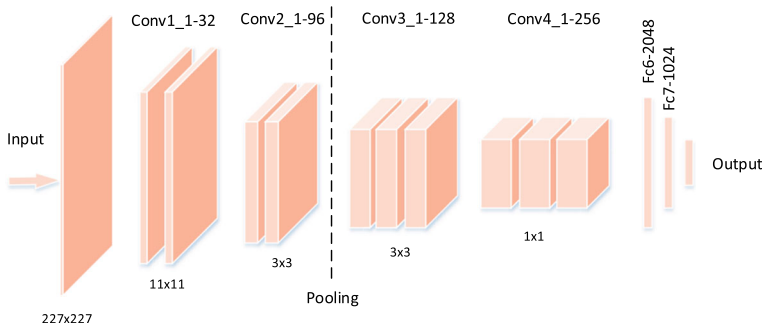

Fig. 3 The second channel of detail-extracted CNN

of insufficient data sets, but also brings a lot of primary characteristics with better generalization performance of smoke. However, we consider that it is not enough to extract the primary features for smoke classification. We add a tidy network to extract the advanced features of smoke. Finally, the features are fused to increase the comprehensiveness of the features and improve the classification accuracy of the model. Moreover, the two channels are trained separately. If one of channels learns features too many or too few, it will not affect the other channel, nor will it directly affect the whole model. So the fusion structure can increase the stability of the model.

The original data sets are input into two channels separately at the beginning. The features of Pool5 in the first channel and Conv4_1 in the second channel are combined depend on the ratio of 1:1. Then these features are put into the next fully-connected layer. After passing through three fully-connected layers, images can be distinguished as smoke and non-smoke images. The structure of DC-CNN is illustrated in Fig. 4.

We crop images to 227 × 227 and manually tag the data sets for training. Two channels are trained separately for data extraction with the same data sets. When passing through the convolutional layer, the input image is convoluted with the convolution kernel and the offset is output. The formula for the convolution principle (1) is as follows:

$$X_j = f\left(\sum_{i \in M_j} X_j \times k_{ij} + \mathrm{b}_j\right) \tag{1}$$

Where M_j represents the input map, l is the number of layers, and k_{ij} is the value of the convolution kernel. Each output map in each layer has a corresponding additional bias b_j. The activation function is ReLU in the experiment.

Each convolutional layer extracts abundant features, which lead to the problem of excessive features and feature dispersion. Regularization is used to reduce and normalize features in t1he proposed model as follows:

$$\overline{X_f} = \frac{1}{m} \sum_{i=1}^{m} X_{i,f} \tag{2}$$

$$\sigma_f^2 = \frac{1}{m} \sum_{i=1}^{m} \left(X_{i,f} - \overline{X_f}\right)^2 \tag{3}$$

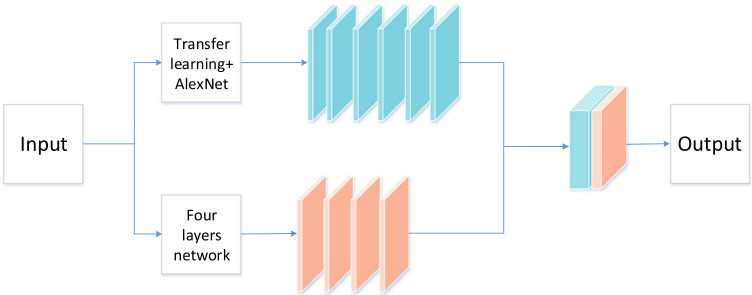

Fig. 4 The structure of DC-CNN

Where m is the size of the small batch, $x_{i,f}$ is the f-th feature of the i-th sample in the small batch. According to the obtained mean (2) and variance (3), the features can be normalized as follows (4):

$$\widehat{X}_f = \frac{X_f - \overline{X_f}}{\sqrt{\sigma_k^2 + \xi}} \tag{4}$$

Where ξ is a small normal number to improve numerical stability. The pooling method in the model is max-pooling, which can reduce the number of parameters and prevent over-fitting.

Particularly, the concat layer is adopted at the end of the dual-channel network. Features are fused by the concat layer with the ratio of 1:1 to splice the two channels. In our model, the concat layer is a splicing on the channel dimension, which is beneficial to have more feature representations and effectively improve the classification performance. So feature maps to be fused must be consistent with N(number), H(height) and W(width). If the channel dimension for each network channel is represented as k1 and k2, the blob output of concat layer can be expressed as: N* (k1 + k2) *H*W [16].

Then input fused features into the next fully-connected layer. Two fully-conneted layers are used to convert feature maps to the size of 1 * n [16]. The eigenvector is converted to smoke and non-smoke output probability by a linear classification module formed by the fully-connected layer(5) and the softmax layer(6):

$$f_{y_i}(X_i; W, b_i) = WX_i + b_i \tag{5}$$

$$F_{y_i}(Z_{y_i}) = \frac{e^{Z_{y_i}}}{\sum_j e^{f_j}} \tag{6}$$

Where x_i,W and b_i are the input variables, the weight matrix and bias. f_{y_i} indicates the function map. F_{y_i} is a fraction of the non-normalized log probability.

4 Experiment

4.1 Data preparation

According to the research of Hongda Tian and Philip O. Ogunbona [24], it is difficult to detect smoke in some challenging images. These challenges can be divided into three categories according to their characteristics: homogeneity, transparency properties and the same physical formation process of smoke. The first category contains vehicle body, clothes, sky and smooth wall due to their image patches have high homogeneity with smoke. The second category includes shadow and glass, which have the same transparency property as smoke. The third category includes some objects sharing similar physical formation process with smoke. The same forming process will make the objects have similar shapes and features, such as cloud, steam, fog and haze. With both transparency property and high homogeneity in some settings, it is very difficult to separate smoke from water.

Fig. 5 Smoke images

In experiment, we construct the data sets based on the public data sets on the website: http://staff.ustc.edu.cn/~yfn/vsd.html [22]. Four challenging images which include clouds in the sky, smooth wall, vehicle body and water, are collected from Internet as supplementary.

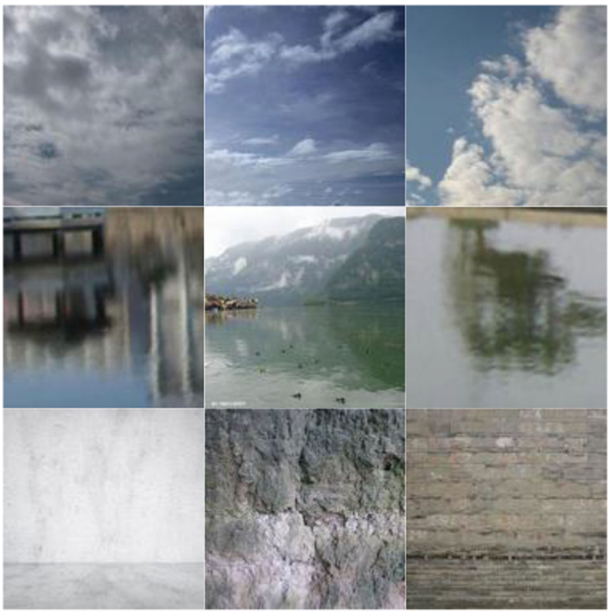

Fig. 6 Non-smoke images

Table 1 The image data sets for training, validation and testing

Datasets	Number of smoke images	Number of non-smoke images	Total number of images	Purposes for
Set1	7934	7934	15,868	Training
Set2	860	860	1720	Validation
Set3	1000	1000	2000	Testing

Samples of smoke images and challenging non-smoke images are shown in Figs. 5 and 6. The details of smoke images and non-smoke images are summed in Table 1. A total of 19,588 pictures are randomly divided into training set Set1, validation set Set2 and test set Set3. The training set occupies the most pictures with 15,868 images. The verification set and test set have 1720 images and 2000 images respectively. The ratio of smoke and non-smoke images in each data set is 1:1.

4.2 Network structure

The proposed network structure consists of two channels. In the first channel, the classical AlexNet network uses transfer learning to initialized weights from the pre-training model. The pre-training model is trained on the ImageNet large-scale data sets. Specifically, the pre-training model is used to initializes the weights of the first convolutional layer (NC1) to the last pooling layer (P5). The hyper-parameter settings are shown in the following Table 2. Since the gradient descent method is used to solve the optimization, it is necessary to set the basic learning rate of 0.001 to prevent over-fitting. The adjustment policy of learning rate is step. The size of step is 1000 and its momentum is 0.9. The weight decay is fixed to 0.0005 to prevent over-fitting. The second channel is a tidy network using to extract more specific features of smoke. Two network are trained together to get the comprehensive features. The final optimized parameters are summarized in Table 3.

4.3 Experimental results

4.3.1 The effectiveness of transfer learning

Transfer learning is used to solve insufficiency of the data sets. By transferring the parameters of the pre-training model, the step of training from scratch is eliminated, and the initialized parameters enable the new model quickly converge to a high level of precision. In order to verify whether transfer learning is suitable to the smoke detection task we make comparative experiments of AlexNet with transfer learning and without transfer learning. The experimental results are shown in Table 4.

The Table 4 shows that, the test accuracy of AlexNet without transfer learning is 0.9696, the loss value is 0.2284. Whereas the AlexNet with using transfer learning increases the accuracy to 0.9908 and the loss value is reduced to 0.0557 in the experiment. The pretraining model

Table 2 Training relevant hyper-parameters

Hyper_parameters	Base_Ir	Ir_policy	Stepsize	Momentum	Weight_decay	Solver_mode
Value	0.001	step	1000	0.9	0.0005	GPU

Table 3 Model parameters maintains

Layer	Layer Type	Hyper-parameters
Input	Input	Image Size:227 × 227
NC1	Normalization and convolution	Filer size k1xk1:11 × 11 Filer number n1:96 Stride s1xs1:4 × 4 Activation function::ReLU
P1	Pooling	Pooling region size k2xk2:3 × 3 Stride s2xs2:2 × 2 Pooling method:max-pooling
NC2	Normalization and convolution	Filer size k3xk3:5 × 5 Filer number n3:256 Stride s3xs3:1 × 1 Activation function::ReLU
P2	Pooling	Pooling region size k4xk4:3 × 3 Stride s4xs4:2 × 2 Pooling method: max-pooling
C3	Convolution	Filer size k5xk5:3 × 3 Filer number n5:384 Stride s5xs5:1 × 1 Activation function::ReLU
C4	Convolution	Filer size k6xk6:3 × 3 Filer number n6:384 Stride s6xs6:1 × 1 Activation function::ReLU
C5	Convolution	Filer size k7xk7:3 × 3 Filer number n7:256 Stride s7xs7:1 × 1 Activation function::ReLU
P5	Pooling	Pooling region size k8xk8:3 × 3 Stride s8xs8:2 × 2 Pooling method:max-pooling
C1_1	Convolution	Filer size k1xk1:11 × 11 Filer number n1:32 Stride s1xs1:4 × 4 Activation function::ReLU
NC2_1	Normalization and convolution	Filer size k2xk2:3 × 3 Filer number n2:96 Stride s2xs2:2 × 2 Activation function::ReLU
Pool2_1	Pooling	Pooling region size k3xk3:3 × 3 Stride s3xs3:2 × 2 Pooling method:max-pooling
C3_1	Convolution	Filer size k4xk4:3 × 3 Filer number n4:128 Stride s4xs4:2 × 2 Activation function::ReLU
C4_1	Convolution	Filer size k5xk5:1 × 1 Filer number n5:256 Stride s5xs5:1 × 1 Activation function::ReLU
Out	Concat	Axis:1
Fc6_1	Fully-connected	Neuron number n6:2048 Dropout probability p:0.5 Activation function::ReLU
Fc7_1	Fully-connected	Neuron number n7:1024 Dropout probability p:0.5 Activation function::ReLU
Output	Output	Neurons number n:2

Table 4 Comparison of testing experimental results

Algorithm	Accuracy	Loss
AlexNet	0.9696	0.2284
AlexNet +transfer learning	0.9908	0.0557

used for transfer learning is trained on Mnist data set, which has a large number of high-quality parameters. Transfer learning transfer these parameters of pretraining model to our model as initialized parameters and the model have superior parameters for convergence from the beginning. The model can achieve a state of rapid convergence and high accuracy. So transfer learning plays an important role in the proposed model. It can accelerate the convergence speed and improve the classification performance of smoke images.

4.3.2 The necessity of two channels

The dual-channel network is proposed to extract more comprehensive features. The first channel is AlexNet using transfer learning, which can get more generalized features. The second channel is trained separately to extract the detailed features of smoke. Finally, the features are fused by the concat layer. These fused features are more comprehensive and the classification performance of the model is optimized by features fusion. We make comparative experiments with two channels and single channel models. The proposed DC-CNN is used as a dual-channel network in the experiment and AlexNet with transfer learning is used as a single-channel network in the experiment. The experimental results are shown in the following Table 5.

As can be seen from the Table 5, the existence of two channels improves the classification performance of the network, and the accuracy value is increased from 0.9908 to 0.9933. In the first channel, the most features parameters obtained by transfer learning are primary. Then we consist a tidy network in the second channel to obtain the advanced features of smoke separately. When the features of the two network channels are fused, the model can obtain more comprehensive smoke features and improve the classification accuracy.Therefore, the transfer learning and the dual-channel network are both helpful for improving classification accuracy in terms of smoke classification.

4.3.3 Comparison experiments

In order to evaluate the classification performance of the proposed model, we compare it against a traditional algorithm and five different deep CNNs algorithms. The traditional algorithm is SC [4]. The first three algorithms adopt LeNet, AlexNet and VGG-16 networks. The third is DNCNN algorithm [29], which has achieved good results in smoke recognition. The forth algorithm is AlexNet network using transfer learning (AlexNet+TL). To adapt to different network structures, we adjust the input images to the size required by each network. Classification accuracy and loss are used to evaluate different deep neural network algorithms.

Table 5 Comparison of two channels and single channel models

Algorithm	Accuracy	Loss
AlexNet +transfer learning	0.9908	0.0557
DC-CNN	0.9933	0.0237

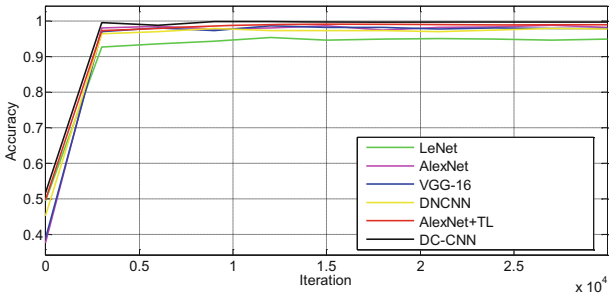

Fig. 7 Accuracy of each network in training

In the training process, from Figs. 7 and 8, we can see that DC-CNN can quickly converge and reach the high classification accuracy. For the loss of DC-CNN, it is lower than other networks at the beginning, and it rapidly reaches a very low loss value. This demonstrates that DC-CNN can converge quickly in the training process.

The experimental test results are presented in Table 6. Firstly, we can see that the classification performance of deep learning algorithms is better than the traditional algorithm obviously. Secondly, among these deep learning algorithms, DC-CNN achieves the highest accuracy and the loss of DC-CNN is the lowest one. It shows that our algorithm achieves the optimum convergence performance. Thirdly, the number of learning parameters of LeNet is 2 M and the number of learning parameters of DNCNN is 43 M, they are less than other networks. The reason is their network input images sizes are 32 × 32 and 48 × 48, whereas input images sizes of other networks are 227 × 227. The size of the input images and the complexity of the network structure will directly affect the number of learning parameters. In conclusion, experimental results demonstrate the distinct advantages of the proposed DC-CNN over the state-of-the-art smoke classification methods.

5 Conclusion

A novel dual-channel convolutional neural network using transfer learning (DC-CNN) is proposed to detect smoke images. The model network takes the AlexNet network with transfer learning as the main channel. The designed detail-extracted CNN serves as the second channel. Two channels are trained separately, and feature fusion is performed at the end of the two channels. The proposed structure can enrich the feature information and ensure the robustness

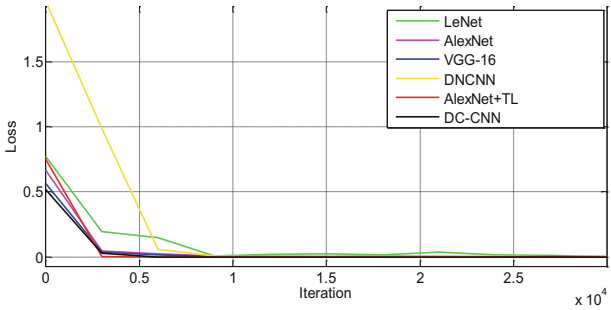

Fig. 8 Loss of each network in training

Table 6 Comparison of testing experimental results

Deep CNNs	Input image size	Accuracy	Loss	Learned parameters
SC [4]	16×16	0.8208	–	–
LeNet	32×32	0.9444	0.2478	2 Million
AlexNet	227×227	0.9696	0.2284	216 M
VGG-16	224×224	0.9754	0.1873	258 M
DNCNN [29]	48×48	0.9826	0.1060	43 M
AlexNet+TL	227×227	0.9908	0.0557	216 M
DC-CNN	227×227	0.9933	0.0238	161 M

of the network. Experimental results show that the proposed model can achieve a better classification performance on the real-world data sets compared with state-of-the-art algorithms.

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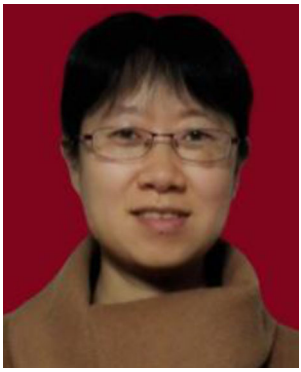

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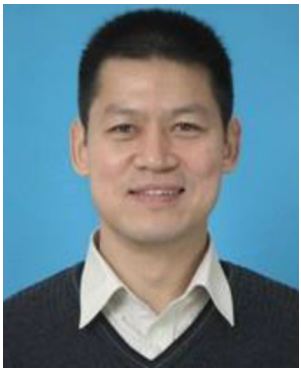

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